

nonparametric statistics

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Kruskal-Wallis Test

```
satisfaction <- data.frame(  
  score = c(7, 8, 5, 5, 7, 6, 5, 4, 8, 9, 6, 5, 8, 9, 8),  
  city = factor(c(rep("CityA", 5), rep("CityB", 5), rep("CityC", 5)))  
)  
  
kruskal.test(score ~ city, data = satisfaction)
```

Kruskal-Wallis rank sum test

data: score by city

Kruskal-Wallis chi-squared = 0.89162, df = 2, p-value = 0.6403

Levene's Test

```
hospital_A <- c(5, 7, 8, 6, 7)
hospital_B <- c(6, 7, 8, 9, 7)
hospital_C <- c(4, 6, 8, 5, 6)
#
data <- data.frame(
  days = c(hospital_A, hospital_B, hospital_C),
  hospital = factor(rep(c("A", "B", "C"), each = 5))
)
#Levene's Test
library(car)
```

Warning: package 'car' was built under R version 4.2.3

Loading required package: carData

```
leveneTest(days ~ hospital, data = data, center = median)
```

```
Levene's Test for Homogeneity of Variance (center = median)
      Df F value Pr(>F)
group  2  0.0833 0.9206
      12
```

Jonckheere-Terpstra Test

```
mild <- c(5, 4, 6, 5, 6)
moderate <- c(6, 7, 6, 7, 8)
severe <- c(8, 9, 8, 10, 11)
#
Data <- data.frame(
  days = c(mild, moderate, severe),
  treatment = factor(rep(c("Mild", "Moderate", "Severe"), each = 5), ordered = TRUE)
)
#Jonckheere-Terpstra
library(DescTools)
```

Warning: package 'DescTools' was built under R version 4.2.3

Attaching package: 'DescTools'

The following object is masked from 'package:car':

Recode

```
JonckheereTerpstraTest(days ~ treatment, data = Data ,alternative = "increasing")
```

Warning in JonckheereTerpstraTest.default(alternative = alternative, c(5, : Sample size > 10
p-value based on normal approximation. Specify nperm for permutation p-value

Jonckheere-Terpstra test

data: days by treatment

JT = 72, p-value = 0.0001337

alternative hypothesis: increasing

Friedman Test

```
scores <- data.frame(  
  student = factor(1:5),  
  exam1 = c(76, 82, 90, 85, 78),  
  exam2 = c(80, 84, 89, 87, 82),  
  exam3 = c(78, 88, 92, 84, 79)  
)  
#  
library(tidyr)
```

Warning: package 'tidyr' was built under R version 4.2.3

```
scores_long <- pivot_longer(scores, cols = starts_with("exam"),  
                             names_to = "exam", values_to = "score")  
#Friedman  
friedman.test(score ~ exam | student, data = scores_long)
```

Friedman rank sum test

data: score and exam and student

Friedman chi-squared = 2.8, df = 2, p-value = 0.2466

Chi-square Test

```
gender_smoke <- matrix(c(30, 10, 20, 40), nrow = 2, byrow = TRUE)
dimnames(gender_smoke) <- list(Gender = c("Male", "Female"), Smoke = c("Yes", "No"))
chisq.test(gender_smoke)
```

Pearson's Chi-squared test with Yates' continuity correction

data: gender_smoke

X-squared = 15.042, df = 1, p-value = 0.0001052

Fisher exact Test

```
small_sample <- matrix(c(2, 7, 8, 5), nrow = 2, byrow = TRUE)
dimnames(small_sample) <- list(Gender = c("Male", "Female"), Smoke = c("Yes", "No"))
fisher.test(small_sample)
```

Fisher's Exact Test for Count Data

data: small_sample

p-value = 0.09907

alternative hypothesis: true odds ratio is not equal to 1

95 percent confidence interval:

0.01410547 1.60171447

sample estimates:

odds ratio

0.194317

McNemar's Test

```
data <- matrix(c(20, 10, 5, 15), nrow = 2, byrow = TRUE)
rownames(data) <- c("Before", "After")
colnames(data) <- c("Support", "Not Support")
#McNemar's Test
mcnemar.test(data)
```

McNemar's Chi-squared test with continuity correction

data: data

McNemar's chi-squared = 1.0667, df = 1, p-value = 0.3017